

African Languages for Detecting Machine Generated Texts

A Comparative Study of Multilingual Models for Detecting Machine-Generated Text in African Languages

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Introduction

The amount of machine generated text circulated exploded with the release of ChatGPT in 2022. Since then, various threats posed by machine generated text have been identified by researchers [1] including:

- Spam and Harassment - Machine generated text simplifies the process of producing harmful social media content by automating it.
- Exploiting AI authorship - machine generated text may allow authors to fake expertise which can be distributed in channels like academia or business.
- Facilitating malware and social engineering - machine generated text may allow digital adversaries to scale operations by using bots to communicate mal-intended information.
- Online influence campaigns - machine generated text can help people fake social media engagement including product reviews and user engagement.

Humans, unfortunately, show little capacity to identify machine generated text [2]. This is why program detectors are important to identifying machine generated text. It is important that access to program detectors is available across a wide variety of languages. That is why this proposal will focus on the creation of detectors for African languages that may not have the level of tools that popular languages like English may have.

Background & Literature Review

Existing work on detecting machine-generated text in low-resource languages has mostly focused on testing how well general AI detection tools perform on African languages, including some South African languages. In many cases, machine-translated text is used as a baseline. Commercial tools like GPTZero and Copyleaks have been evaluated on texts translated from English into Southern Sotho and isiZulu. Although Copyleaks has performed better than some other detectors in certain cases, its reliability still needs improvement. For example, one study found that GPTZero failed to recognise any of the machine-translated texts in Southern Sotho and isiZulu as machine-generated [5].

In recent years, research has started to move towards building more comprehensive models and datasets for African languages. One important development is the MultiSocial dataset, which covers 22 languages across five social media platforms [3]. This dataset gives researchers a way to test detection models on shorter, more informal texts that are closer to the kind of language people use on social media.

There has also been progress in developing Afri-centric transformer models. AfroXLMR, along with its social media-specific version AfroXLMR-Social, was pretrained on 76 African languages, including South African languages. These models have shown that domain-adaptive pretraining (DAPT) and task-adaptive pretraining (TAPT) can improve performance on subjective NLP tasks like sentiment analysis, emotion detection and hate speech classification in these languages [4].

Research has also shown that training models on multiple languages improves cross-lingual generalisation [3]. For example, multilingual models often perform better than monolingual ones when classifying data from low-resource languages, especially when there is only a limited amount of training data available.

Research Questions

Existing studies show that commercial detectors do not always perform reliably on translated African language text, while newer African-focused models have mainly been developed for related NLP tasks rather than machine-generated text detection itself. There is therefore still limited understanding of how well pretrained multilingual models can detect machine-generated text in these languages, whether transfer learning improves performance in lower-resource settings, and how well such models generalise across languages, domains and text generators.

This project is guided by the following research questions:

1. How accurately can human-written and machine-generated text be distinguished in selected African languages?
2. Which pretrained multilingual model performs best for this task, and does transfer learning improve performance in lower-resource languages?
3. How well do these models generalise across languages, domains and different LLM generators?

Dataset & Data Description

This project will use the NCHLT Text Corpora as the main source of human-written text for Afrikaans, English and isiZulu. The corpus is publicly available through the SADiLaR Corpus Portal and contains about 2.57 million Afrikaans words, 12.76 million English words, and 1.97 million isiZulu words drawn mainly from South African government-domain documents [7].

Machine-generated texts will be collected separately by prompting selected LLMs, producing a labelled dataset for binary classification between human and machine-generated text.

Approach & Methodology

This project will model the task as a binary text classification problem, where each text in Afrikaans, English and isiZulu is labelled as either human-written or machine-generated. Human written samples will be drawn from the NCHLT Text Corpora, and machine-generated text will be produced by prompting these multiple LLMs of varying size and architecture to paraphrase the human written source texts [3]. We will first establish baseline performance using TF-IDF with logistic regression, TF-IDF and SVM, and a character n-gram classifier. These methods are suitable because they are computationally efficient and able to capture surface-level patterns and lexical features that may differentiate human and machine-written text. We expect that these will provide a good benchmark, though they might struggle when machine-generated text is more natural and fluent.

The main experiments will compare AfriBERTa, AfroXLMR and XLM-R, with AfroXLMR as the main model for transfer learning. These multilingual transformer models are appropriate because they can learn contextual representations across languages. We expect that the transformer models will outperform the baselines, and that AfroXLMR will perform best because it is more closely aligned with African languages. To keep the scope feasible, only the strongest two models will be used for generalisation experiments across languages and different LLM generators.

Evaluation Strategy

Evaluation of the resulting program will start with metrics quoted within [6] as part of standard machine learning evaluation. These include:

1. Precision - $\text{true detections of machine text} \div \text{total predictions of machine text}$
2. Recall - $\text{detections of machine text} \div \text{total instances of machine text}$
3. F-score - $(2 * \text{precision} * \text{recall}) \div (\text{precision} + \text{recall})$
4. Accuracy - $\% \text{ of machine text identified}$

The robustness of the program will be evaluated by charting the standard metrics across various sources of text for comparison. The metrics will be used to inform an explanatory analysis of the program's functions across all sources.

Expected Outputs & Contributions

We expect fine-tuned multilingual transformer models to outperform traditional or general-purpose baselines for machine-generated text detection in selected African languages, since prior work shows that fine-tuned detectors generally perform better than zero-shot or pretrained baselines in multilingual settings [3]. We also expect performance to vary across languages, mainly because of differences in training data availability, language similarity, and writing style. Previous work on African-language models highlights data imbalance across languages as an important challenge [4]. Finally, we expect machine-generated text to display patterns such as repetition, lower lexical variety, and more formulaic phrasing, but that these cues will not be identical across all languages. Instead, explainability analysis is likely to show that useful detection signals differ by language, domain and generator [3].

References

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